

Lab Worksheet 3: Pseudocode

CS 101 Algorithmic Problem Solving

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1. Rooks Attack

You are given a standard 8×8 chessboard which has exactly 2 rooks on it and no other pieces. The rows are numbered 1 to 8 from bottom to top, and the columns are numbered 1 to 8 from left to right. The cell at the intersection of the i -th column and j -th row is denoted (i, j) .

Given the initial positions of the rooks in the form of coordinates (x_1, y_1) and (x_2, y_2) . you need to tell whether the 2 rooks currently attack each other or not. Assume, each square can contain at most one piece.

Hint: Rooks can only travel in straight lines along the row or column they are placed at, and can't jump over other pieces.

Constraints

- $1 \leq x_1, x_2, y_1, y_2 \leq 8$
- $(x_1, y_1) \neq (x_2, y_2)$

Interaction

The input comprises a single line containing 4 space-separated integers denoting the values of x_1, y_1, x_2 and y_2 respectively.

The output must contain a single line stating Yes if the rooks attack each other, and No otherwise.

Sample

Input	Output
1 2 5 2	Yes
1 1 8 8	No

In the first case, The two rooks can attack each other by moving along the second column.

In the second case, No matter how a rook moves it cannot reach the second rook in one move. Hence, they do not attack each other.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
1 2 1 5	Yes
2 4 4 8	No
3 3 4 3	Yes

Pseudocode

Input: X1,X2,Y1,Y2

Output:

if X1==X2 or Y1==Y2:

 print("Yes")

else:

 print("No")

Dry Run

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: X1=2, Y1=4, X2= 4, Y2= 8

if 2==4 or 4==8:(both are false)

 print("Yes")

else: (code will execute this)

 Print("No")

Output = No

Input: X1=3, Y1=3, X2=4, Y2=3

if 3==4 or 3==3:(true, code will execute this)

 print("Yes")

else:

 print("No")

Output = Yes

2. Where's the lamb biryani?

Michelin starred chef and lamb sauce enthusiast, Gordon Ramsey, wants to learn how to make a perfect biryani. For this reason, he's come to Karachi to learn under the tutelage of Uncle Farhan, owner of the renowned Farhan Biryani restaurant.

Uncle Farhan charges X rupees per class. Chef Ramsey can attend Y classes per week. Uncle Farhan suggests he should attend classes for N weeks in order to become a true master.

Given values of X , Y , and N , determine the total amount of money Chef Ramsey will have to pay.

Constraints

- $X, Y, N \in \mathbb{N}$
- $1 \leq X, N \leq 10^5$
- $1 \leq Y \leq 7$

Interaction

The input comprises a single line containing 3 space-separated integers denoting the values of X , Y , and N respectively.

The output must contain a single number denoting the total amount of money Chef Ramsey has to pay.

Sample

Input	Output
500 3 3	4500
1000 7 1	7000

In the first case, $(X, Y, N) = (500, 3, 3)$. Uncle Farhan charges 500 rupees per class and Chef Ramsey has to take a total of 9 classes (3 classes per week for 3 weeks). He has to pay a total of $500 \cdot 9 = 4500$ rupees.

In the second case, $(X, Y, N) = (1000, 7, 1)$. Uncle Farhan charges 1000 rupees per class and Chef Ramsey has to take a total of 7 classes (7 classes per week for 1 week). He has to pay a total of $1000 \cdot 7 = 7000$ rupees.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
1350 5 4	27000
500 7 52	182,000
1337 3 10	40,110

Pseudocode

Input: X, Y, N

Output:

```
totalcost=X*Y*N  
print(totalcost)
```

Dry Run

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: X=500, Y=7, N=52

```
totalcost=500*7*52  
print(totalcost)
```

Output = 182,000

Input: X=1337 Y=3 N=10

```
totalcost=1337*3*10  
print(totalcost)
```

Output=40,110

3. Mood Lift

There are three special dishes that Chef can make at her restaurant. When a customer eats any of these, their mood elevates by an amount equal to A , B or C , depending on the dish. Chef's favorite customer is in a foul mood and Chef wants to give his mood a boost. But she only has time to prepare any 2 of the 3 dishes.

Given A , B , and C , determine the maximum boost that Chef can provide.

Constraints

- A, B, C
- $1 \leq A, B, C \leq 10^8$

Interaction

The input comprises a single line containing 3 space separated integers denoting the values of A, B, C respectively.

The output should contain a single integer denoting the maximum boost.

Sample

Input	Output
4 2 8	12
10 10 100	110

In the first case, $(A, B, C) = (4, 2, 8)$. The maximum boost is $8 + 4 = 12$.

In the second case, $(A, B, C) = (10, 10, 100)$. The maximum boost is $100 + 10 = 110$.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
18 271 31	302
127 2933 182	3115
21 2 18	39

Pseudocode

Input: A, B, C

Output:

```
n=0
if A>=B>=C and B>=A>=C:
    print(A+B)
elif A>=C>=B and C>=A>=B:
    print(A+C)
else:
    print(B+C)
```

Dry Run

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: A=21, B=2, C=18

```
if 21>=2>=18 or 2>=21>=18: (both false)
    print(A+B)
elif 21>=18>=2 and 18>=21>=2:(true)
    print(A+C)
else:
    print(B+C)
```

Output=39

Input: A=127, B=2993, C=182

```
if 127>=2993>=182 or 2993>=127>=182:
    (both false) print("A+B")
elif 127>=182>=2993 or 182>=127>=2993:
    (both false) print(A+C)
else:
    print(B+C)
    (2993+182)
```

Output=3115

4. Weapon Division

A warlord has been terribly injured after a battle and it's looking like his time on Earth is coming to an end. Before he departs from the world, he wants to distribute all K of his weapons somewhat fairly among his N sons.

Unfortunately, his math skills have deteriorated greatly after decades of constant warring and bloodshed, and he has asked you for assistance in distributing the weapons. He has set just one condition: the difference between the son that got the least weapons and the son that got the most weapons should be no greater than 1.

Given values of K and N , determine the **minimum** number of weapons a son can have if all K weapons are distributed among them according to the above condition.

Constraints

- $K, N \in \mathbb{N}$
- $1 \leq K \leq 10^9$
- $1 \leq N \leq 10^5$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the value of K and N respectively.

The output must contain a single number denoting the **minimum** number of weapons a son can possess once all K weapons are distributed according to the warlord's condition.

Sample

Input	Output
8 5	1
5 8	0

In the first case, $(K, N) = (8, 5)$. The warlord has 8 weapons which are to be distributed among 5 sons. The first 5 weapons can be distributed equally among the 5 sons. The remaining 3 weapons will then be given to any 3 sons. Therefore, 3 sons will receive 2 weapons and 2 sons will receive 1 weapon. Hence, minimum number of weapons possessed by a son is 1.

In the second case, $(K, N) = (5, 8)$. The warlord has 5 weapons which are to be distributed among 8 sons. The 5 weapons can be given to any 5 sons equally. The remaining 3 sons will receive no weapons. Hence, the minimum number of weapons possessed by a son is 0.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
15 5	3
39 10	3
24 25	0

Pseudocode

Input: K, N

Output:

```
answer=K//N  
print(answer)
```

Dry Run

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: K=15, N=5

```
answer=15//5  
print(answer)
```

Output=3

Input: K=39, N=10

```
answer=39//10  
print(answer)
```

Output=3

5. Grading

HackerLand University has the following grading policy:

- Every student receives a grade in the inclusive range from 0 to 100.
- Any grade less than 40 is a failing grade.

Sam is a professor at the university and likes to round each student's grades according to these rules:

- If the difference between the grade and the next multiple of 5 is less than 3, round grade up to the next multiple of 5.

- If the value of grade is less than 38, no rounding occurs as the result will still be a failing grade.

Given the grade g of Sam's student, find out the final grade after rounding as per his rules.

Constraints

- $0 \leq g \leq 100$

Interaction

The input comprises a single line containing an integer denoting the value of g .

The output must contain a single number denoting final grade after rounding.

Sample

Input	Output
84	85
57	57

In the first case, $g = 84$ and the next multiple of 5 is 85, $85 - 84 = 1$ which is lesser than 3. Therefore, it is rounded to the next multiple of 5 which is 85.

In the second case, $g = 57$ and the next multiple of 5 is 60, $60 - 57 = 3$ which is not lesser than 3. Therefore, it is not rounded and remains as is.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
29	29
78	80
51	51

Pseudocode

Input: g

Output:

$R = g \% 5$

```
if  $g \leq 38$ :
    print( $g$ )
elif  $R \geq 3$ :
    print( $g + (5 - R)$ )
else:
    print( $g$ )
```

Dry Run

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input g=78
R=g%5

if 78<=38: (false)
 print(g)
elif R>=3: (true)
 print(g+(5-R))
else:
 print(g)
Output=80

Input g=51
R=g%5

if 51<=38: (false)
 print(g)
elif R>=3: (false)
 print(g+(5-R))
else:
 print(g)
Output=51

6. Siege Truck

Arpasland is surrounded by attackers. A truck enters the city. The driver claims the load is food and medicine from Iranians. Ali is one of the soldiers in Arpasland. He doubts about the truck, maybe it's from the siege. He knows that a tag is valid if the sum of every two consecutive digits of it is even.

Determine if the tag of the truck is valid or not.

Constraints

- Tag can not contain more than 5 digits.

Interaction

The input comprises a single line containing a 5 digits number representing the tag.

The output must be a Valid or Non-Valid depending on the validity of the tag based on the condition mentioned above.

Sample

Input	Output
12446	Invalid
35119	Valid

In the first case, sum of the first and second digit is odd hence, it is invalid.

In the second case, sum of all the consecutive digits is even although the digits are odd hence it is a valid tag.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
24311	Invalid
64088	Valid
97316	Invalid

Pseudocode

```

Input: tag
c1=tag%10
tag=tag//10
c2=tag%10
tag=tag//10
c3=tag%10
tag=tag//10
c4=tag%10
c5=tag//10
if (c1+c2)%2==0 and (c2+c3)%2==0 and (c3+c4)%2==0 and (c4+c5)%2==0
    print("Valid")
else:
    print("Invalid")

```

Dry Run

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: code=24311

```

c1=24311%10
tag=24311//10
c2=tag%10
tag=2431//10
c3=tag%10
tag=243//10
c4=tag%10
c5=24//10
if (1+1)%2==0 and (1+3)%2==0 and
(3+4)%2==0 and (4+2)%2==0 (false)
    print("Valid")
else:
    print("Invalid")

```

Output= Invalid

Input:code= 64088

```

c1=tag%10
tag=64088//10
c2=tag%10
tag=6408//10
c3=tag%10
tag=640//10
c4=tag%10
c5=64//10
if (8+8)%2==0 and (8+0)%2==0 and (0+4)%2==0 and
(4+6)%2==0 (TRUE)
    print("Valid")
else:
    print("Invalid")

```

Output= Valid

7. Icebreaker

Lee is a part of a karate demonstration and his next task is to break three blocks of ice with widths W_1 , W_2 , and W_3 stacked on top of each other (W_1 is on the top and W_3 is on the bottom).

Lee's punch has a strength of S . Whenever he strikes at the stacked blocks, consider the largest $k \geq 0$ such that the sum of the width of the topmost k blocks does not exceed S ; the topmost k blocks shatter and are removed from the stack. Lee can also choose to reverse the order in which the blocks are stacked before every hit, without incurring any penalty.

Given values of W_1 , W_2 , W_3 , and S , find the minimum number of hits Lee needs to break through all three blocks if he performs the reversals optimally.

Constraints

- $W_1, W_2, W_3, S \in \mathbb{N}$
- $1 \leq W_1, W_2, W_3 \leq S$
- $1 \leq S \leq 10$

Interaction

The input comprises a single line containing 4 space-separated integers denoting the value of W_1 , W_2 , W_3 , and S .

The output must be a single integer denoting the minimum number of hits required to break all three blocks if the reversals are performed optimally.

Sample

Input	Output
1 2 2 3	2
1 1 1 3	1

In the first case, $(W_1, W_2, W_3, S) = (1, 2, 2, 3)$. Lee can break through the first two blocks in one hit as $W_1 + W_2 \leq S$ and then the last block in a subsequent hit. Alternatively, he could flip the order, break through W_3 and then break through the remaining two blocks in the next hit. Minimum number of hits required is 2.

In the second case, $(W_1, W_2, W_3, S) = (1, 1, 1, 3)$. Lee can break through the three blocks in one hit as $W_1 + W_2 + W_3 \leq S$. Minimum number of hits required is 3

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
5 1 5 6	2
7 5 6 10	3
4 4 1 9	1

Pseudocode

Input: W_1, W_2, W_3, S

Output:

if $W_1 + W_2 + W_3 \leq S$:

 print(1)

elif $W_1 + W_2 \leq S$ or $W_2 + W_3 \leq S$:

 print(2)

else:

 print(3)

Dry Run

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: 7, 5, 6, 10

```
if 7+5+6<=10:
    print("1") (FALSE)
elif 7+5<=10 or 5+6<=10:
    print("2") (FALSE)
else:
    print("3")
```

Output=3

Input: 4, 4, 1, 9

```
if 4+4+1<=9:
    print("1") (TRUE)
elif 4+4<=9 or 4+1<=9:
    print("2")
else:
    print("3")
```

Output=1

8. Fitness Fun

In order to promote physical fitness, Liebling University is holding a trek along trails of length 100 km, 210 km and 420 km. Students who complete any trek within D days will win a prize of amount A, B , and C respectively ($A < B < C$). There can be multiple winners for a trek and a student can participate in only one trek.

Given the distance, d km, that Chef can cover in a single day, find the maximum prize that she can win at the end of D days.

Constraints

- D, d, A, B, C
- $1 \leq D \leq 25$
- $1 \leq d \leq 500$
- $1 \leq A < B < C \leq 100000$

Interaction

The input contains a single line containing five space-separated integers denoting the values of D, d, A, B , and C respectively.

The output must contain an integer indicating the maximum prize that Chef can win. The output must be 0 if no prize can be won.

Sample

Input	Output
10 5 100 200 300	0
20 15 15000 20000 30000	20000

In the first case, $(D, d, A, B, C) = (10, 5, 100, 200, 300)$. The distance covered by Chef in 10 days is $10 \times 5 = 50$ km which is less than any of the available distance categories. Hence Chef won't be able to claim any prize.

In the second case, $(D, d, A, B, C) = (20, 15, 15000, 20000, 30000)$. The distance covered by Chef in 20 days is $20 \times 15 = 300$ km which satisfies the first and second treks but not the third. Hence Chef can claim a maximum prize of 20000.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
10 9 1100 2400 3600	0
10 20 2000 3000 4000	2000
15 25 2000 3000 4000	3000

Proposed Pseudocode

```

distance = D*d
if distance >= 100 and distance < 210:
    return A
elif distance >= 310 and distance < 350:
    return B
elif distance >= 420:
    return C
else:
    return 0

```

Dry Run

Using any two of the inputs provided in the Exercise section above, dry run the proposed pseudocode in the space below.

Input: 10, 20, 2000, 3000, 4000
distance=10*20
if distance>=100 and distance<210: (TRUE)
return A
elif distance>=310 and distance<350:
return B
elif distance>=420:
return C
else:
return 0
Output= A=2000

Input 15, 25, 2000, 3000, 4000
distance=15*25
if distance>=100 and distance<210: (FALSE)
return A
elif distance>=310 and distance<350:(FALSE)
return B
elif distance>=420:(False)
return C
else:
return 0

Error Identification

Output= 0 (expected output=3000)

Briefly explain any errors that you identified in the proposed pseudocode. Mention the line number(s) and the errors in each line.

The range in the forth line is wrong, it should be from distance>=201 to distance<420

Correct Solution

Below, rewrite the lines of code that you mentioned above with their errors corrected.

Input: D, d, A, B, C
Output:
if distance>=100 and distance<210: (TRUE)
return A
elif distance>=210 and distance<420:
return B
elif distance>=420:
return C
else:
return 0

9. Favorite Numbers

Alice likes numbers which are even, and are a multiple of 7. Bob likes numbers which are odd, and are a multiple of 9. Alice, Bob, and Charlie find a number A .

- If Alice likes A , Alice takes home the number.
- If Bob likes A , Bob takes home the number.
- If both Alice and Bob don't like the number, Charlie takes it home.

Given A , find who takes it home.

Constraints

- $1 \leq A \leq 1000$

Interaction

The input comprises a single line containing an integer denoting the value of A .

The output must contain the name of the person who takes A home. Name shall be as the following: Alice, Bob or Charlie.

Sample

Input	Output
7	Charlie
27	Bob

In the first case, 7 is not even, hence Alice doesn't like it. It is odd, but isn't a multiple of 9. Hence Bob doesn't like it. Therefore, Charlie takes it home.

In the second case, 27 is odd and a multiple of 9. Therefore, Bob likes it and takes it home.

Proposed Solution

```

if (A % 2 != 0) and (A % 7 == 0):
    print("Alice")
elif (A % 2 != 0) or (A % 9 == 0):
    print("Bob")
else:
    print("Charlie")

```

Dry Run

Using any two of the inputs provided in the Sample section above, dry run the proposed pseudocode in the space below.

Input: 27

```

if (27%2 != 0) and (A%7 ==0): (False)
    print("Alice")
elif (A%2 != 0) or (A%9 == 0): (True)
    print("Bob")
else:
    print("Charlie")

```

Output= Bob

input: 7

```

if (7%2 != 0) and (7%7 ==0): (True)
    print("Alice")
elif (7%2 != 0) or (7%9 == 0): (False)
    print("Bob")
else:
    print("Charlie")

```

Ouput= Alice

Error Identification

Briefly explain the errors you identified in the proposed code solution. Mention the line number and errors in each line.

In the first line not equals "!=" is written, instead equality "==" should be used.

In the third line "or" is used, where "and" should be used

Correct Solution

Rewrite the lines of code you mentioned above with their errors corrected.

Input: A

Output:

```
if (A%2 == 0) and (A%7 ==0): (False)
    print("Alice")
elif (A%2 != 0) and (A%9 == 0): (True)
    print("Bob")
else:
    print("Charlie")
```